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D1.2 Demarcating collaborative awareness from related concepts

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¹R: Document, report (excluding the periodic and final reports); DEM: Demonstrator, pilot, prototype, plan designs; DEC: Websites, patents filing, press & media actions, videos, etc.; DATA: Data sets, microdata, etc.; DMP: Data management plan; ETHICS: Deliverables related to ethics issues.; SECURITY: Deliverables related to security issues; OTHER: Software, technical diagram, algorithms, models, etc.

²PU – Public, fully open, e.g. web (Deliverables flagged as public will be automatically published in CORDIS project's page); SEN – Sensitive, limited under the conditions of the Grant Agreement; Classified R-UE/EU-R – EU RESTRICTED under the Commission Decision No2015/444; Classified C-UE/EU-C – EU CONFIDENTIAL under the Commission Decision No2015/444; Classified S-UE/EU-S – EU SECRET under the Commission Decision No2015/444

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Abstract

Awareness in biological agents has converging definitions when considering local states describing content-related consciousness from an agent-specific perspective. However, it becomes highly debated when it comes to global states. The issue magnifies when considering collectives of artificial agents. Several frameworks exist, all unsatisfactory in the limitations posed to agents' heterogeneity and disappearance of the local self into an integrated state.

Ultimately, existing frameworks are ineffective in explaining, facilitating, and supporting cooperative behaviours in artificial agents. The lack of a compelling theory of global awareness in AI is currently a significant barrier to the effective deployment of artificial agents in the real world.

EMERGE tackles this grand challenge by introducing the novel concept of collaborative awareness for collectives of minimal artificial beings. We will investigate how simple agents can develop a representation of their mutual existence, environment, and cooperative behaviour towards the realisation of tasks and goals.

EMERGE builds on a scenario of artificial beings with no shared language and constrained individual capabilities, which nevertheless leads to high-complexity behaviours at the collective level. Collaborative awareness becomes an emergent process supporting complex, distributed, and loosely coupled systems capable of high degrees of collaboration, self-regulation, and interoperability without predefined protocols.

EMERGE delivers a philosophical, mathematical, and technological framework that enables us to know how and where to allocate awareness to optimally achieve a goal through the collective. We will demonstrate EMERGE concepts on robotic use cases, with hints of the broader applicability of the framework to the Internet of Things, pervasive computing, and nanotechnologies. We will also investigate the ethical implications of collaborative awareness, focusing on moral responsibility, vulnerabilities, and trust.

Consortium

The EMERGE consortium members are listed below.

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Università di Pisa	UNIPi	IT
TU Delft	TUD	NL
University of Bristol	UOB	UK
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Executive Summary

This document is a deliverable of the EMERGE project, funded under grant agreement number 101070918.

This deliverable, “D.1.2 Demarcating collaborative awareness from related concepts”, is reports on the activities of WP1 “Conceptual framework” which aims to elaborate and refine a new framework for collaborative awareness tailored for a set of heterogenous agents.

In this document, we review a number of concepts and frameworks that could accommodate a form of collective awareness in systems powered by artificial intelligence (AI), and extend to human agents.

In particular, we outline what a proper framework needs to account for, and how a new notion of collaborative awareness can meet its requirements. Building on and advancing beyond existing notions such as Central Integrated Awareness (CIA) and Distributed Situational Awareness (DSA), collaborative awareness introduces a new understanding of awareness in collectives. Unlike CIA, which centralizes the perception and processing of information, or DSA, which emphasizes the dispersed nature of awareness across different agents, collaborative awareness advocates for an interconnected approach that leverages the strengths of both integration and distribution.

1. Introduction

We are deeply entrenched in an expansionist phase of AI developments, with new systems and applications being constantly developed and refined. A new challenge however is emerging: as domain-specific AI systems are getting more numerous but also more diverse, sometimes because of their association with particular architectures, brands or even geographical areas, their coexistence is becoming increasingly problematic. As we enter this second phase of AI evolution, the pressure is not only to master and adapt to new environments, but share the same ones with others — sometimes very different — “species” of AI, as well as with us, humans (see Deroy et al., under review)

The concerns we commonly hear about - a car's inability to navigate a flooded road or a drone's hesitation between two equally perilous obstacles - are serious. But a more serious one, which attracts less attention, is the risk of automated vehicles or drones from different manufacturers colliding due to a lack of coordination in the spaces that they will de facto share. This situation not only presents immediate safety risks but also poses a deeper problem: the increasing difficulty for humans to comprehend not just the functioning of individual systems but also their collective operation and interaction with people. If our assessment is accurate, the paramount challenge of the next generation of AI systems lies in ensuring the coexistence or coordination of multiple task-specific systems in shared spaces, many of which will be occupied by humans as collaborators, controllers, users, or parallel occupants (Bacciu et al., 2021).

To solve the later part of the challenge, it seems essential to understand how humans or existing living beings interact with each other. When it comes to humans, most of our theoretical understanding comes from rather small group interactions, for instance mundane situations with two people solving a coordination problem in an economic game, or moving a large box. These situations usually presupposes two kinds of mental states, local and mutual. To focus on the box example only, the local states for each person include visual awareness

of the box and of the other agent, the proprioceptive awareness of one's body, the spatial awareness of the box in the room, the awareness of how to exercise pressure on the box to keep a grip on it, etc. Local states may be hard to enumerate or classify but they are at least relatively easy to grasp. The mutual states are harder to pinpoint but no less essential : in the box moving example, each of the two agents needs to coordinate their intention and be aware that the other is doing the same to cooperatively move the box to the centre of the room. Like other human-like concepts, we should ask whether the concepts of mutual awareness and cooperation offer the right sources of inspiration for AI systems.

More importantly for large collectives of AI systems, the mutual awareness model may be too demanding, in at least four respects. The first one, already mentioned, is that **mutual awareness does not necessarily scale up to large groups**. It is possible to infer the intentions of one or two people, but if one has to coordinate more, or large ensembles, independent resources (planning tools, coordinators) are needed to ensure the coordination or cooperation across agents.

Second, mutual awareness models of cooperation often require either verbal communication and trust (one of the agents explicitly commit to work on a shared goal, and the other trust them to do so) and a rich and somewhat continuous form of non-verbal exchanges that convey meaning or attention (the agents facial expressions, gaze, gestures and signs communicate the constant engagement in the task, and help to solve coordination problems on a moment to moment basis). These **mentalist and social concepts do not necessarily have analog in AI agents, nor should they have** (Deroy, 2022 for trust).

Third, human cooperation is — at least in many accounts — supported by **acculturation and social or moral norms** (e.g Göckeritz et al, 2014 for early experimental evidence ; Schmidt and Rakoczy, 2023 for recent review, see also Dorsch & Deroy, accepted) which come with special forms of social and cultural learning, which so far have little to do with the kind of learning implemented in AI agents.

Fourth, while examples of mutual awareness discussed in the literature tend to show successful coordination, the reliance on mutual mind-reading and inference of intentions also open the door to some failures of coordination, while other, simpler forms of coordination and/or cooperation may be more robust.

Collective awareness (sometimes studied as “collective experience” in humans) also remains more broadly problematic, partly because we still lack consensus about what it amounts to in biological agents, including humans: collective intelligence (as sharing of information) can be modelled and measured, but the awareness which exists for a collective, remains controversial or understood under different frameworks such as plural self-awareness, collective intentionality, distributed situational awareness, or we-mode cognition.

An additional difficulty here is to see that most forms of collective awareness in biological agents, if any, are meant to take place amongst homogeneous agents (that is to say, within the same conspecifics). Collaborative awareness between aware artificial agents need to include possibly heterogeneous capabilities.

Our focus in EMERGE is therefore to define a form of artificial awareness, distinct from the forms of awareness at stake in human cooperation, which can **safely, efficiently and ethically enable large scale collaboration between individual agents, including heterogeneous ones**.

What kind of awareness - which we call collaborative awareness - can act as a solution for AI-AI collaboration? Can it accommodate humans' practical and ethical interests in their interaction with groups of AI?

2. Background terminology

As already specified in D1.1, we consider awareness as distinct from consciousness, which entails the occurrence of a phenomenal experience (qualia) and an essentially private, unique and unshareable point of view.

We interchangeably use the terms **unitary**, **singular**, or **individual AI agent/system** to refer to an agent that strives to attain its objective as a single unit.

In contrast, we distinguish two kinds of collective agents : we use the term collective agent to designate a set of AI agents, while we use **collaborative AI agent/system** to refer to a set of agents that maximally strive to collaborate in a space or on a task.

It is important to stress that collaboration can occur between agents of the same kind (two humans, two bots) or different kinds (one human, one bot ; one bot of type A and one bot type B). We refer to homogenous collectives when those are made of similar AI agents, and heterogeneous when they are made of distinct systems.

3. What do we mean by collaboration ?

3.1. The need for a bespoke concept

In the same way that we need to conceptually engineer a new definition of awareness for artificial agents, we need to do the same for collaboration when it comes to the collective behaviour and decisions between AI or between AI and humans.

Cooperation is a topic that has garnered attention from researchers across various fields, with some focusing on human behaviours such as psychology, sociology, anthropology, and economics, and others examining interactions among non-human organisms, including ethology, behavioural ecology, and evolutionary ecology. Philosophers and ethicists also have taken a great interest in the social dimensions of decisions, either in small-scale cases of joint action and social cognition, or broader issues having to do with promoting the common good.

There are two main reasons to move slightly away from the concept of cooperation, according to us.

- a. While theoretical and empirical perspectives on cooperation constitute a fertile ground and source of inspiration for the field of AI (e.g. Duéñez-Guzmán et al, 2023.), we also insist that interactions between living beings or human-like concepts may limit the range of solutions that can and should be explored with artificial systems. We therefore opt for redefining or conceptually engineering the concept of collaboration for our purposes, and recognise the possible overlap with certain accounts of cooperation or differences with existing use of the term “collaboration” in other fields (see Box 1, on the method).
- b. The terminology used across various fields is not agreed upon. This is particularly the case for words like cooperation and joint action.

For some, ‘cooperation’ is a certain type of interaction that can be recognised by form (e.g. participants take turns receiving benefits, participants synchronise their behaviours) and/or

outcome (participants renounce individual benefits for collective ones; e.g. Bowles & Gintis (2003, pp. 429–430). Economists define cooperation as ‘an individual behaviour that incurs personal costs to engage in a joint activity that confers benefits exceeding those costs to other members of one’s group’. This is how many biologists would define ‘altruism’. For some, it is a strategy used by a participant in an interaction, and for others it is a characteristic style of a long-term relationship. For instance, according to Bronstein (2003), an ecologist with expertise in interspecific interactions, cooperation, mutualism, and symbiosis are defined as distinct categories. Cooperation refers to beneficial interactions between individuals of the same species. In contrast, mutualism and symbiosis describe beneficial relationships between individuals of different species. Symbiosis encompasses all types of close and long-term interactions, while mutualism is specific to short-term interactions.

While cooperation tends to be studied across multiple trials or occurrences, joint actions are studied when focusing on a single event, where two or more agents coordinate their plans of action to achieve an external outcome together (Sebanz, Bekkering, & Knoblich, 2006). Again the terminology is not fixed, but joint actions can be isolated, for instance when two agents jointly agree to take the same train, and never cooperate after that.

Many theories highlight the importance of certain foundational elements in joint action, though there is a diversity of views regarding these essential aspects. These foundational concepts include collective goals (Miller 2001), irreducible joint intentions (Searle 1990, 1995), reasons for action constructed collectively (Tuomela 2007), normative commitments (Gilbert 1989), and mutually consistent plans of action (Bratman 1992, 1993). The list provided here is not exhaustive, but suffices to illustrate the diversity of views present in this field.

Box 1. Extending conceptual engineering for AI

Conceptual engineering is a growing area in the philosophy of language aimed at improving the concepts we use. It goes beyond just looking at how concepts are built or relate to each other, aiming instead to improve, change, or create new concepts to better meet our needs. At its core, conceptual engineering acknowledges that many of our concepts are flawed or limited. They might be vague, ambiguous, or misaligned with our values and goals. For instance, concepts like "truth," "knowledge," "gender" or "information" are central to various philosophical, scientific, and social discourses, yet they are often contested or problematic.

Conceptual engineering represents a proactive approach to philosophical analysis. It emphasises the practical implications of conceptual analysis and the potential for philosophical inquiry to contribute to real-world improvements. By examining and revising the building blocks of our thought and communication, conceptual engineering aims to make our concepts more precise, effective, and aligned with our ethical and epistemological goals. This evolving field highlights the dynamic nature of philosophical inquiry and its relevance to contemporary challenges.

Prominent figures in conceptual engineering include David Chalmers (2020), who has discussed the methodology and scope of conceptual engineering, and Herman Cappelen (Burgess, Cappelen & Plunkett, 2020), who has critically examined its foundations and implications. Their work, along with others like Amie Thomasson (2021), who has contributed to the understanding of how concepts function within our linguistic practices, demonstrates the field's diverse applications, from metaphysics to social philosophy.

In EMERGE, and in the methods we support, this idea of conceptual engineering plays a role, though our motivations and practices are different in at least three core respects :

- For AI, the issue might be more about missing the right concepts rather than fixing outdated ones. We need to focus on making sure the new terms can help with designing new and better systems, rather than match existing issues.
- Revised or bespoke concepts should be intelligible and easy to adopt for the public (Deroy, 2023) which is using these systems, as a matter of ethics of communication.
- Revised or bespoke concepts should be operationalised in computer science and robotics.

3.2. Three-levels of collaboration

We define **collaboration** as the time-bound adaptive pursuit of multi-agent goals, including through changes in environment and group states or composition.

Each term is defined below.

(i) **Collaboration** encompasses a spectrum of interactions between agents, which can be categorised into three distinct degrees: communication, coordination, and shared decision-making. Each level represents a deeper and more integrated form of collaboration, ranging from simple information exchange to collective action. Each level stresses a different meaning of “multi-agent goals”.

0- Co-existence goals: Communication is a core aspect of collaboration outside the presence of established norms⁴ and differs from broadcasting and is responsive to successful reception in the collective(s). At this stage, the primary focus is on the exchange of information between coexisting agents. There's no imperative for action or decision-making. For example, in a workplace setting, this might involve team members informing each other about their progress on individual tasks through emails or meetings. The key is that while everyone is informed, no collective decision or coordinated action is required based on the information shared. Communication is collaborative in that it is not just a passive event : The function is to make one's state or information transparent or visible to others in a certain appropriate context, where the act of communication is itself meaningful. Communication can happen through messages or transfer of information, or through the environment (placing a brick in the environment, sending a message, touch).

1- Interdependent goals: Coordination elevates the collaboration to a level where the actions or decisions of the agents are interdependent, though they may still pursue distinct individual goals. This requires not just communication but also a deliberate process to align actions in a way that ensures non-interference and global efficiency. An example of coordination could be seen in a group project, where team members divide tasks among themselves. Each member works on a different aspect of the project, but they adjust their schedules and efforts to ensure the project comes together cohesively and meets deadlines.

⁴Coordination around focal points that people are aware should be salient is a case where coordination can occur without communication.

Coordination involves planning and adjusting actions in response to others' needs and contributions, even as each participant may have a separate goal.

2- Common goals: Cooperation represents the highest degree of collaboration, where agents' decisions and actions are unified towards a common goal. At this level, individual decisions and actions are considered collectively, which means that the lines between personal and group objectives are blurred. This can be exemplified by a team of firemen during a rescue operation: every action is decided upon a shared goal (extinguish the fire) and pursued according to a collective mode of operation. Though the team operates with a single goal, each member's action contributes to the team decision-making process. The outcome usually (but not necessarily) could not have been attained by a single member. Another example would be that of a symphonic orchestra.

In collective robotics, an alternative characterisation comes from considering super-additivity in performance (Figure 1 centre and right). In general, coordination tends to not be super-additive while cooperation tends to be super-additive.

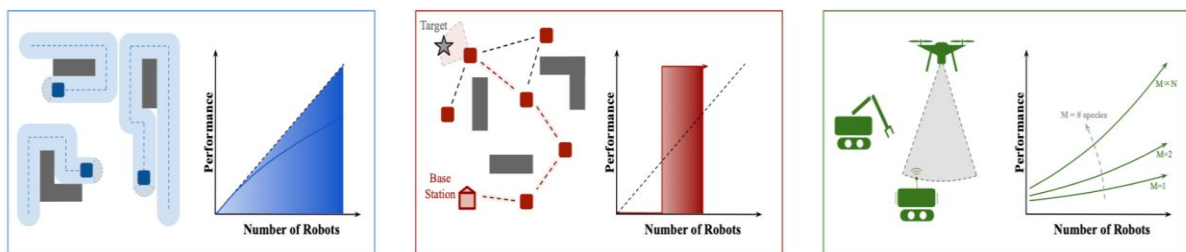


Figure 1 (from Prorok et al.2021). Linear interactions (left) vs. superadditive interactions between homogenous (middle) and heterogeneous (right) agents. Super-additive interactions mean that the gain in performance increases with the number of agents, though sometimes only after communication is established.

Case Study 1

We can apply the three degrees of collaboration—communication, coordination, and shared decision-making—to the context of logistics robots working in a warehouse environment. These examples will illustrate how robots can work individually and collectively to achieve efficiency and productivity in logistics operations.

Communication

At the communication level, logistics robots are equipped with the ability to exchange information about their status, location, and tasks without necessarily coordinating their actions. For instance, a fleet of robots in a warehouse might share their positions and battery levels with a central system (in case of central control) or with each other (in case of swarms). No collective decision or action is taken based on this communication; it simply serves to keep the robots informed about the operational environment, and eventually share this information with other robots (cleaning robots for instance) or humans (users, controllers).

Coordination

Moving to coordination, logistics robots not only share information but also adjust their actions based on the actions and needs of other robots to achieve individual tasks efficiently. An example of this would be robots programmed to handle different stages of the packing and shipping process. One robot picks items from inventory and places them on a conveyor belt, another packs these items into boxes, and a third prepares the packed boxes for shipping. Each robot's actions are adjusted based on the status and output of the others—speeding up

or slowing down to match the flow of items and ensuring that there's no bottleneck in the process. This requires more than mere communication; it involves an integrated system where robots' tasks are interdependent, though their goals (completing their specific task) remain distinct. This coordination again can include humans.

Cooperation

At the highest level of collaboration, shared decision-making, logistics robots work together towards a common goal, making decisions as a unified entity. In this scenario, robots in a warehouse are not only aware of each other's positions and tasks but also collaborate to decide on the most efficient way to organize and execute work based on real-time demand and operational conditions. For example, if a sudden surge in orders occurs, the robots collectively decide how to reallocate tasks among themselves to meet the increased demand efficiently. This might involve some robots temporarily shifting from their usual tasks to assist in packing and shipping orders. The decision on how to redistribute the workload is made collectively, considering the capabilities and current status of each robot to optimize overall performance. This level of collaboration ensures that the group of robots operates as a cohesive unit, adapting dynamically to changes and optimizing workflow in a way that individual robots or simple coordination could not achieve. Shared decision-making can also include humans in different roles.

Case Study 2

We can also apply the three degrees of collaboration—communication, coordination, and shared decision-making—to a soft robot, designed for instance like an octopus with various semi-independent arms. Each arm of the soft robot can function with a certain degree of autonomy, yet all are part of a cohesive unit, simulating the interconnected yet independent operation seen in biological octopuses.

Communication

At the communication level, each semi-independent arm of the soft octopus robot has sensors that gather information about the environment, such as object location, texture, and temperature. Part of this information can be shared across all arms. For instance, if one arm detects an object that requires handling, it communicates this information to the other arms. However, at this stage, there is no coordinated effort to act upon this information together. The primary purpose is to ensure that each arm is aware of the environment and the activities of other arms.

Coordination

In the coordination phase, the soft robot's arms start to adjust their actions in response to the information shared and the activities of other arms, working towards individual tasks with an underlying sense of collective efficiency. For example, if the robot is tasked with assembling a product, one arm might pick up a component and position it, while another arm, recognizing the action of the first, fetches the next component needed for assembly. This requires the arms to not only communicate but also coordinate their actions by timing their movements to avoid interference and ensure a smooth workflow. The coordination is evident as the arms work on separate tasks but in a manner that reflects a mutual awareness and a shared operational strategy.

Cooperation

At the highest collaboration level, shared decision-making, the semi-independent arms of the soft robot operate with a unified purpose, making collective decisions to achieve a common

goal efficiently. This can be demonstrated in a scenario where the robot needs to navigate through a complex, confined space to perform maintenance or repairs. The arms collectively decide how to distribute the tasks of locomotion, obstacle manipulation, and tool usage among themselves to navigate and complete the task effectively. This might involve some arms focusing on gripping and moving objects out of the way, while others concentrate on propelling the robot forward or using tools. The decision on how to approach the task is made collectively by the robot's central processing unit, considering input from all arms to optimize movement and task execution. The arms function as extensions of a single entity, with each action being a result of collective decision-making rather than individual or merely coordinated efforts.

Case Study 3

More tentatively, we suggest that these three concepts of communication, coordination, and shared decision-making can also be applied to Artificial Neural Networks (ANNs)

Communication

At the communication level in ANNs, the analogy can be seen in the forward propagation of inputs through the layers of the network. Each neuron in a layer receives inputs, processes them, and then communicates its output to neurons in the subsequent layer. This process does not involve any coordinated action or decision-making across neurons concerning the overall task; instead, it's a straightforward pass-through of information. The neurons are simply relaying information based on their individual activations, akin to agents sharing information without a collective goal.

Coordination

Coordination within ANNs can be viewed in the context of backpropagation and optimization algorithms, where the network adjusts the weights of the connections between neurons based on the error or loss calculated at the output. This process requires a form of coordination where neurons adjust their behaviour (i.e., how they weigh inputs) based on the collective goal of reducing overall error. Although each neuron's adjustment is based on local gradients and the architecture's overall structure, there's an implicit coordination aiming to align individual neuron adjustments towards improving the network's performance on a task. This coordinated effort ensures that the network learns to map inputs to outputs more accurately over time.

Cooperation

Shared decision-making in the context of ANNs involves ensemble learning, where multiple ANNs might work together, each making predictions or decisions that are then combined through voting, averaging, or another method to produce a final decision. Here, the "shared decision-making" comes from aggregating the outputs of multiple networks to arrive at a decision or prediction that reflects a consensus among them.

(ii) **Time-bound:** Tasks or objectives are often set within a specific timeframe, which necessitates efficient collaboration and management of resources. Being time-bound means that collectives have a definite time frame to achieve their goals within the allocated period.

Time-bound collaboration differs from more open-ended cooperation, which lacks a fixed deadline. Open-ended cooperation holds more often within communities and in-groups and focuses on continuous improvement, and mutual support. The primary objective is to cultivate sustainable, long-term benefits rather than achieving immediate, time-sensitive goals.

(iii) **Adaptive pursuit:** We need to highlight the dynamic nature of collaboration, where collectives must be flexible and responsive to changes. Adaptability is key in navigating

unforeseen challenges, adjusting strategies, or incorporating new information. The adaptive pursuit ensures that collaboration remains effective even when conditions or requirements evolve.

We consider adaptation as covering both changes in the environment and in group states or composition. In other words, the external and internal conditions affecting collaboration can shift.

Within our case studies, we can imagine a warehouse setting where a collective of bots is assigned the task of moving boxes from various incoming locations to designated storage areas. The warehouse environment is subject to change – new shipments may arrive unexpectedly, storage zones could be reassigned, or obstacles might emerge in the bots' paths due to ongoing operations. When an unforeseen obstacle blocks the predefined path, the bots must quickly reassess their routes. They communicate among themselves to identify and disseminate information about the obstacle, then calculate alternative routes to their destinations without human intervention. Or suppose a new, heavier type of box arrives, requiring more bots to safely transport it than initially anticipated. The swarm must adapt by reallocating sufficient bots to handle the heavier load, ensuring safety and efficiency without explicit reprogramming by human operators.

Changes in the environment can combine with changes in the group. The changes of environment mentioned above might include reassigning roles within the collective to ensure that bots are utilised where they're most needed, based on their current location, battery life, or carrying capacity. If some bots are taken out of service for maintenance or experience malfunctions, the remaining units must reorganise their tasks and formations to maintain operational efficiency.

We can also consider the case of a soft robot in a research lab, designed to assist with a variety of tasks such as handling delicate biological samples. The lab environment is constantly changing: new experiments are set up, equipment is moved, and tasks can vary day by day. Upon receiving a new, fragile sample type or a piece of equipment that the soft robot hasn't encountered before, the arms must adjust their grip strength and manipulation strategy. If one arm has reached its operational limits, the robot's control system reallocates tasks to other arms, ensuring that the workload is distributed evenly. This internal reorganisation allows the robot to continue its activities without interruption, optimising productivity and extending the operational life of each arm. For tasks requiring more precision or force than a single arm can provide, the robot's arms work in concert, adjusting their movements and force application in real-time. This collaborative effort is based on the current state of each arm and the specific requirements of the task, ensuring that delicate operations are performed safely and efficiently. In this case, the soft robot, much like the swarm of bots, exemplifies the need to dynamically respond to both external environmental changes and internal changes in its capabilities and states.

At the same time, the collective awareness that needs to be provided to AI differs from the case with biological agents, as it can be engineered into the set of agents to enable their coordinated action, regulation and control. What makes this a form of collaborative awareness is that it is irreducible to the sum of local states.

4. Critical review of existing concepts and models

4.1. Emergent collective behaviour

The phenomenon of emergent collective behaviour among living organisms shows that group-level behaviour does not need to rely on the reading of intentions or predictions (Couzin, 2008). It allows decentralised groups to rapidly detect and respond to environmental threats and opportunities. The mere collecting and aggregating of information, which can be implemented in a variety of ways, can allow groups notable advantages (Bettencourt, 2009).

Bacterial quorum sensing is a prime example of emergent collective behaviour. Bacteria use quorum sensing to coordinate gene expression according to the density of their population. This process involves the production, release, and subsequent detection of signalling molecules called autoinducers. When the concentration of these molecules reaches a **certain threshold**, it triggers a coordinated response from the bacterial community, such as biofilm formation, virulence factor production, or bioluminescence. This allows bacteria to undertake complex tasks that are beyond the capabilities of individual cells, such as evading host immune responses or optimising resource utilisation (Whiteley et al., 2017).

This model is already labelled as a form of biological swarming (Sharma and Anand, 2002) and inspires some AI swarming strategies (see for instance Chandrasekara and Hougen, 2006; Martinez, 2020).

It is important to see that however in this quorum sensing awareness is missing because they don't "know" that they will reach or have reached quorum. This is the difference that is missed when comparing humans organised in close feedback loops and swarms (as in Rosenberg et al., 2016)

Flocking behaviour is found in groups of birds, insects, and fish, and showcases the ability of individual agents to move in unison, thereby forming a dynamic and cohesive system (for instance in birds: Emlen, 1952; Ballerini et al, 2008). This collective behaviour, deeply rooted in the principles of self-organisation and emergent properties, allows these groups to exhibit complex movements and formations without a central control mechanism, relying instead on simple rules of alignment, cohesion, and separation among individuals. Research in the field of ethology and computational biology has provided insights into how these decentralised systems achieve such high levels of coordination, emphasising the role of local interactions between individuals within their immediate vicinity.

The study of flocking behaviour has also inspired the development of artificial systems that mimic these biological models (e.g. Vásárhelyi et al., for a representative example). By replicating these dynamics, engineers and scientists aim to create robotic agents capable of performing complex tasks collaboratively, enhancing the system's overall efficiency and responsiveness. Designing such behaviours in artificial systems necessitates that each robotic agent aligns its movement strategy—adjusting orientation and speed in relation with fellow agents. This involves recognizing and locating other agents and obstacles, determining their relative positions, and coordinating movement to prevent collisions, preserve system integrity, and fulfil assigned tasks.

Collective spatial searches are another example of emergent collective behaviour present in a range of organisms. Beyond flocks of birds and schools of fish, this phenomena is often focused on the organisation of social insect colonies (Bettencourt, 2009). Generally, this group behaviour takes the shape of a large number of individuals all setting out from a single starting point (their nest or hive), and exploring their environment to find food or new nest locations. When this reward has been located a signal will be given to indicate to others where this can

be found. In ants this signal takes the shape of a pheromone trail, the ant that returns with food the quickest will leave the strongest track - thereby leading others to preferentially opt for the same path (Hölldobler & Wilson, 1990). Bees provide this signal through what has been termed a 'waggle dance' which provides the orientation, and the duration of the dance is indicative of the quality of the reward (Seeley, 2010). With numerous bees performing such dances, the strength of a specific location will gain prominence, and through quorum sensing the swarm of bees will (in most cases) arrive at a consensus.

Numerous algorithms have been modelled based on this collective behaviour in insect colonies. One such being Ant Colony Optimization (ACO) which is designed for application to various continuous optimization problems, among which AI (Blum, 2005; Dorigo, Birattari & Stutzle, 2006; Dorigo & Stutzle, 2018). The behaviour in bees has also given rise to a tool for optimization problems, the Bees Algorithm (BA), which has also found application in swarm robotics (Pham et al, 2006; Pham & Castellani, 2009).

Table 1 - A comparison between biological and artificial swarms (from Duan et al, 2023)

	Biological swarm	Artificial swarm
Distribution	No central node, only local interactions	No central command, only local interactions
Simplicity	Simple perception and behaviour rules	Low cost simple rules
Flexibility	Adapts to environmental changes; avoids threats	Adaptive to environmental changes; many tasks
Intelligence	Emergent	Emergent

Collective maze learning by rats represents another unique example of emergent collective behaviour. Research has demonstrated that rat groups can navigate mazes more efficiently than solitary rats. This collective learning mechanism relies not on the knowledge or experience of any individual rat but on the interactions among several rats exploring the maze together. Through vocalisations and other signals, they share information about successful routes and dead ends, collectively enhancing their maze navigation strategy.

Researchers (Nagy et al, 2020) utilised automated video tracking to observe that rats, even when alone, engage in a partially systematic search. This behaviour gradually improves their chances of locating a reward due to a growing attraction to unexplored areas. In group settings, however, synergistic advantages emerge from combining individual exploration and social interactions. The enhanced search performances stem from a strategy that prioritises responses to social and asocial cues in a hierarchical manner. Additionally, the study introduces a computational model to examine the key factors influencing collective search dynamics and confirms that the **collective search strategy boosts individual search efficiency in group contexts**.

Three primary factors dictate tunnel-choice decisions: rats are more likely to move (1) toward a tunnel that was visited longer ago (78% versus 22%), (2) in a direction leading to more unvisited endpoints (60% versus 40%), and (3) towards visible conspecifics (63% versus 37%). The study unveils a decision-making hierarchy among rats that facilitates a systematic

search pattern, significantly enhancing the likelihood of locating a target in a maze as the number of unvisited endpoints increases. This hierarchy, which balances the value of prior experience (i.e., exploring unvisited parts of the maze) against social information (i.e., moving toward group members), renders the group more efficient than an equivalent number of isolated individuals. Despite relatively local interactions among individuals—owing to the maze's design, which mimics natural tunnel systems and limits long-distance visual signal transmission—a 'follow others' tactic combined with the avoidance of previously visited tunnels yields synergistic benefits for the group. However, as simulation results indicate, highly efficient searches only materialise when individuals display both exploratory and social behaviours, as reliance on either behaviour alone leads to suboptimal performance.

This approach may offer valuable insights for developing computational search algorithms and systems, such as autonomous robot groups.

Collective decisions in social animals are notably intricate, often encompassing the participation of numerous individuals, ranging from a few to thousands. Notable instances are observed among social insects like honey bees and ants, as well as in fish, and mammals including primates, meerkats, African wild dogs, bison, and deer (for overviews see: Arganda et al, 2012 Seeley & Buhrman, 1999; Conradt & Roper, 2005; Conradt & List, 2009).

Collective decisions emerge from a collective where each has unique needs and insights into a complex environment ; they are also characterised by significant conflict of interests, where erroneous choices could lead to fatal outcomes. Such conflicts often arise from the varied necessities of individuals, such as seen in primate groups (e.g. King & Sueur, 2011; Sueur & Deneubourg, 2011), or from exclusive knowledge about distinct locations, as observed in swarming honeybees (Seeley, 2010).

Due to the high cost associated with information gathering, animals frequently depend on their peers to become informed. This leads to a form of communication within social species, where signals of information and needs are shared, facilitating a “collective deliberation” process that evaluates the significance of each signal as an indicator of individual intent. This dynamic process of collective decision-making sees a rotation in the identities of individuals contributing information or expressing needs, ensuring diverse participation in setting agendas and deliberations. The random variation in participant identities across decisions, driven by stochastic factors in physiological or information-gathering processes, introduces a selection mechanism based on randomness rather than formal selection.

While there is evidence for both voting and copying mechanisms in animal systems, differentiating between these two mechanisms is challenging, not least because they can produce similar individual and group-level behaviours. Indeed, distinguishing between voting and copying mechanisms in the wild requires that researchers continuously and simultaneously measure group members' orientation and movement towards a target destination, which is logistically challenging (e.g. in free-ranging goats: Sankey et al, 2021). Genuine voting behaviours are meant to involve intentional signalling and, arguably, an understanding of the mental states of others (It's crucial to note that even when animals may engage in collective decision-making, the cognitive processes they employ differ fundamentally from those in humans).

4.2. The human case : Mutual awareness and collective responsibility

Following Searle's (1990) famous examples of people running to a shelter together or merely simultaneously, philosophers have tended to agree that whatever joint action or cooperation may be, it is underdetermined by observable behaviour and depends on internal states.

Accordingly, a general trend of the literature consists in defining joint intentions, which are supposed to be to joint actions what individual intentions are to individual actions (for an overview see Schweikard and Schmidt, 2021).

Based on human cases, joint action is often thought to require a shared understanding and continuous monitoring of each other's intentions, actions, and outcomes (Sebanz & Knoblich, 2021). Awareness in this context involves several layers: self-awareness, awareness of others, and mutual awareness. Self-awareness is crucial as it allows an individual to recognise their role and responsibilities within the joint action, or social group at large (Zhao, 2018, Focquaert & Platek, 2006). It ensures that each participant can adjust their actions based on the ongoing dynamics of the interaction, maintaining for instance the alignment of their contributions with the collective objective in the case of a full collaboration.

Awareness of others is equally important. It involves understanding the capabilities, intentions, and actions of other participants. This understanding enables individuals to anticipate the actions of their partners, facilitating seamless coordination. During interactions, individuals not only observe and react to partner actions but they also need to predict the actions of their partner and their expected consequences (Sebanz & Knoblich, 2009).

Mutual awareness, the collective understanding among participants, is however what often truly characterises human joint action. It emerges from the continuous interaction and communication between individuals, allowing them to adjust their behaviour in real-time. This mutual understanding ensures that all participants are aligned towards the common goal, even as situations change. Besides verbal communication, gestures and eye contact are often used to maintain and adjust mutual awareness during joint actions.

Two different directions offer key theoretical frameworks to think of collaboration without verbal communication.

In philosophy of cognitive science, Steven Butterfill (e.g. 2016), alongside his colleagues, has proposed that simpler forms of joint action do not necessitate complex cognitive processes such as shared intentions or sophisticated theory of mind capabilities. His conceptual framework for minimal joint action highlights at least three key features. First, minimal joint actions are characterised by a form of joint commitment among the participants towards a shared goal. However, this commitment does not require the participants to have a mutual understanding or shared intentions in the full philosophical sense. Instead, it requires a more basic level of coordination and cooperation, and simply memory of rules. Second, the actions are directed towards collective goals that can be understood and pursued even without sophisticated mental state attributions. This implies that participants in a minimal joint action are oriented towards a common outcome but do not necessarily need to have a detailed plan or agreement on how to achieve it. Finally, a key element of minimal joint action is reciprocal responsiveness, where participants adjust their actions in response to the actions of others to achieve the collective goal. This responsiveness does not require complex communication or understanding of others' intentions but is based on a more immediate and direct form of coordination.

Game theory offers a framework for analysing and computing the control mechanisms at play in multi-agents scenarios. Within collaborative endeavours, the perceived benefit for each participant is influenced by the condition of all involved parties. A game typically involves at least two players whose interests may partially overlap or diverge. In certain collaborative situations, participants might establish a consensus on a unified strategy through direct communication before the action, thereby acting in unison. However, there are cases where no pre-action consensus on strategy is formed. In these instances, coordination naturally evolves as participants gather insights into their counterparts' actions, the consequences

thereof, and potentially their ultimate objectives, employing a variety of mechanisms through ongoing or repeated interactions. In scenarios with conflicting objectives, coordination develops incrementally as participants gradually understand the dynamics of their interaction, task demands, and each other's actions.

4.3. Central integrated awareness and distributed awareness in AI

Shared awareness across many AI agents enables the coordination and shared understanding among multiple AI systems in order for them to work together towards common goals or to manage tasks in a distributed environment. By providing an extra-level for sharing information and insights, the agents can collectively achieve a level of understanding and responsiveness that surpasses what they could achieve individually. However, there are significant challenges and limitations. Ensuring seamless communication and understanding among multiple agents becomes increasingly difficult as the number of agents grows. This can lead to coordination overhead, where the resources and time spent managing interactions between agents detract from their overall effectiveness. Privacy and security concerns are amplified in such systems, as the sharing of information across agents increases the risk of data breaches or malicious exploitation.

An alternative framework has been advocated, based on distributed awareness. The concept derives from that of distributed cognition, a theoretical framework in cognitive science that posits that cognitive processes are not confined solely to the individual but are distributed across objects, individuals, artefacts, and tools in the environment. For instance, when a pilot navigates an aeroplane, cognition is seen as distributed across the pilot, the cockpit instruments, the procedures, and the air traffic control system. There, cognition extends beyond the boundaries of the individual to include the surrounding environment, social interactions, and technology.

The Distributed Situation Awareness (DSA) framework, initially introduced by Stanton et al. in 2006 also challenges traditional, individual-centric views of Situation Awareness (SA), such as those proposed by Endsley in 1995, by presenting SA as a property emerging from the collective interactions within socio-technical systems, encompassing both human and technological agents. Like in situated cognition, this perspective posits that SA does not solely reside within the cognitive realms of individuals but is a systemic attribute manifesting through the dynamics between different system components. Illustrating this concept, Salmon et al. in 2018 explored DSA within road intersection scenarios, highlighting how different participants, including drivers, motorcyclists, pedestrians, traffic management systems, and advanced vehicle technologies, possess unique perspectives and contribute varied information to a shared DSA network. This ecosystem of perspectives ensures the safe operation of the intersection by facilitating the exchange and integration of relevant information among all participants, ensuring that each agent possesses an awareness that, while not necessarily identical, is compatible with that of others in the system.

4.4. Hierarchical learning

In the field of artificial intelligence research, there has been significant exploration into synchronising the goals of various components or sub-agents within a system. Methods such as leveraging population dynamics for equitable reward distribution or encouraging diverse incentives for cooperation have proven effective. These strategies suggest the practicality of overarching frameworks in achieving alignment of objectives at different levels.

Hierarchical reinforcement learning offers an additional method by breaking down intricate problems into manageable layers. In this structure, lower-level controllers are tasked with specific sub-tasks, while higher-level controllers manage the selection and continuation of these tasks. This tiered system consolidates different strategies under a single supervisory mechanism, enhancing the efficiency of the learning process throughout the system. Nonetheless, the integration of learning processes between higher and lower-level controllers poses certain challenges. To successfully implement specialised controllers, it is crucial for them to excel in their designated tasks and to be in sync with the system's broad objectives. Moreover, these controllers require constructive feedback on their performance within the wider system context to improve their strategies.

The challenge also includes the adaptability of the lower-level units. The complexity of these units can affect the overall system efficiency; more advanced units may boost system performance, whereas simpler units might be more readily steered towards desired outcomes. Employing a strategy of simplification may facilitate coordination but at the expense of diminishing the system's capacity for innovation and adaptation to novel information.

5. Dimensions and capacities for emergent collaborative awareness

5.1. Collaborative awareness: A dimensional model

Collaborative awareness is not strictly modelled on human or animal collaboration. It only implies a capacity to **share one's current situation on a specific dimension** (i.e. what to share : for instance time, distance to goal, distance to others, energy-levels, expected energy-expenditures, confidence in sensory detection) with other agents, added to various filters (i.e. who/when to share) and simple collaboration-protocols (communication, coordination, cooperation). These protocols frame the dimensions required to achieve collaborative awareness, in three different levels: communicative awareness, coordinated awareness, and cooperative awareness.

Reaching the level of communicative awareness only minimally extends the existing dimensions of individual awareness - for instance, a communicatively aware fleet of drones or logistic bots can be aware of members' location, energy levels, current task-achievements. Uncertainty levels can also be communicated.

In a mutual communicative awareness, a working solution is to rephrase the dimensions present in individuals and add new dimensions as "common ground for spatial awareness", "common ground for temporal awareness" etc.

Coordinated awareness also replicates all dimensions but has at least some necessary additional components: one needs to have dimensions supporting a form of "social awareness" keeping track of each unit or group independent goals and plans, separately from one's own goals and plans.

Cooperative awareness requires in turn additional dimensions - mainly the awareness of the common goal. Here we anticipate that other dimensions will be useful, though not necessary, such as a way to compute "mutual uncertainty" or "common ground" between the agents as to how their decisions contribute to these shared goals.

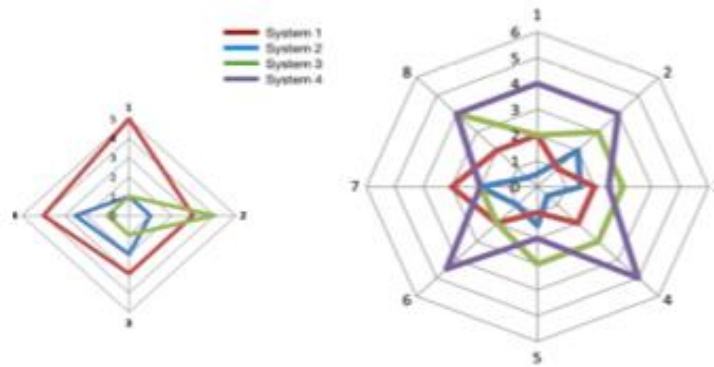


Figure 2 - Dimensional model of the local agents (left) and their emergent integration in collaborative awareness (right)

The benefits of collaborative awareness are threefold:

1. The focus is on specific collaborative tasks, and specific dimensions of awareness. This means that the contribution of awareness to a system can be measured and compared towards engineering the most appropriate awareness.
2. By contrast with the dominant views of consciousness, collaborative awareness then needs not to be integrated or unified: one system may need to share some aspects with another system, or with their controller, but share other aspects with other systems or their users.
3. Collaborative awareness can be partitioned, scaled and most importantly shared. Because it can be instantaneously or continuously shared between agents, collaborative awareness is radically different from the “private theatre” of consciousness which we enjoy and struggle with understanding in others. It resembles more a selective telepathy.

Our understanding of emergent collaborative awareness is anchored in two aspects: it acknowledges a dependence on local awareness while also acknowledging the comparative autonomy of emergent collaborative awareness. Our perspective embraces the philosophical idea of strong emergence, which delineates dependence through nomological or causal connections, meaning that emergent dimensions go beyond their underlying local dimensions. Central to this model are two principles:

- (1) Non-aggregativity: Rather than being mere compilations of their lower-level counterparts, emergent properties manifest through novel, self-organising behaviour that could not be possible or similar without them.
- (2) Distinctive efficacy: Emergent collaborative dimensions wield unique causal abilities that cannot be diminished to the dimensions of their local elements.

5.2. Concepts for implementation of cooperative strategies

How can cooperation, the higher level of collaboration, be achieved? The question remains a major challenge and we outline four different directions pursued as part of our project.

(A) Space: Continuity and proximity

Spatial continuity and proximity can act as trigger for communication, enabling units to collaborate more dynamically and responsively. This spatial model already facilitates real-time data exchange and decision-making processes in swarms.

(B) Similarity

Collaborative awareness can be helped by developing communication protocols that prioritise and facilitate interactions among units with shared goals (or identities, e.g. brand affiliations). Adaptive learning mechanisms can be implemented here to enable units to adapt their cooperation strategies based on outcomes, further aligning their actions with the shared goals.

(C) Team reasoning

Team reasoning is a specific approach that is interested in a fundamental shift in decision-making from a self-centred "I" perspective to a collective "We" mindset, where the individual's considerations are expanded to include the group's interests (Bacharach 1999; Sugden 1993, 2015; Colman and Gold 2018; Karpus and Radzvilas 2018). This transition involves an alteration in the individual's agency, or the way the individual construes it, though the specifics of this change and its ontological implications are often not clearly defined.

The essence of team reasoning is captured in a straightforward framework, notwithstanding subtle (though also important) variations across different interpretations. This framework posits that: (1) an individual recognizes themselves as part of a team; (2) there's mutual understanding among team members that each identifies with the team; (3) all team members agree on the goal of maximising a utility function that reflects the team's interests; (4) it is universally acknowledged that a specific set of actions optimally maximises this utility function, and each member is expected to play their part in this action set.

In contrast to individualistic reasoning, where decisions are made by all interacting parties in isolation, considering the potential outcomes based solely on the individuals' personal gain without regard for the impact on others, team reasoning introduces a collective perspective. It implies a mental alignment among team members, viewing their decisions and outcomes as interconnected, fostering a cooperative approach that addresses coordination challenges by aligning individual actions with the team's overarching goals.

The concept that understanding others' preferences goes beyond merely anticipating their actions predates the formal development of team reasoning theory (Radzvilas and Karpus 2021). This notion is subtly embedded in a widely accepted Nash equilibrium refinement within traditional game theory, aimed at resolving decision-making dilemmas in scenarios presenting multiple equilibria in interactive decision situations. One such refinement, payoff dominance, suggests that players will gravitate towards the equilibrium that offers the best outcome for all parties involved. This implies that individuals consider not only their personal preferences but also those of others in their decision-making process (Schelling 1980; Harsanyi and Selten 1988). The adoption of the payoff-dominant strategy, therefore, is not just a predictive assumption but a reflection of a deeper consideration of mutual best interests, indicating a shift towards a more integrated and cooperative decision-making framework.

(C) Sharing uncertainty

A string of studies undertaken by EMERGE members (Bahrami et al 2010; Pescetelli et al 2015; Navajas et al, 2022) have stressed the importance of communicating uncertainty in collective decisions: groups can significantly enhance decision-making by incorporating members' self-assessed confidence levels. EMERGE's approach is strongly shaped by these findings.

Those previous studies demonstrated that group performance can be notably improved when members communicate not only their choices but also their confidence levels in these choices. Confidence acts as an indicator of the reliability of each member's contribution, enabling more effective aggregation of information across members by weighting them according to confidence. Through sharing of uncertainties, the group can effectively weigh each member's input, granting more influence to those with higher confidence which is presumed to correlate with accuracy, thus aiding in information aggregation to boost the likelihood of correct decisions - when sensitivities are matched.

Further studies and findings have shown that the approach is applicable across various contexts (Kurvers et al, 2023; Dezechache et al, 2022), suggesting that from scientific teams to medical teams, any group can improve decision-making by fostering the sharing of confidence levels. Social learning is also important in refining group decision-making capabilities over time, as members become better at interpreting and leveraging communicated uncertainties, indicating a dynamic improvement in decision accuracy.

6. The Human-Collaborators in the loop

Implementing collaborative awareness in a collective of AI is also important for reasons that extend beyond the mere coordination of actions to encompass ethical, legal, and social dimensions.

Mutual awareness plays a role for the interpretation and evaluation of the group action. Individual voluntary human actions entail a certain degree of sense of agency, that is, the feeling of voluntarily controlling one's own actions and their consequences in the outside world (Haggard, 2017). In joint actions, and as a result of their coordinated efforts, agents may experience a sense of shared control over the outcome they achieved together. Philosophers propose that this feeling of shared control over joint actions and their external outcomes may be defined as "sense of joint agency" (Pacherie, 2012). The reason to insist on this aspect is the interplay between the sense of agency and the attribution of responsibility, a crucial aspect in moral and legal accountability. It posits that for an individual to be held responsible, they must have had control over their actions and a sense of intentionality behind those actions.

This is also the case for collective actions. Thus, the awareness involved in a collective sense of agency does not only contribute to the feeling of being in control but also grounds the ethical and legal principles that determine responsibility for groups.

This concept of "sense of joint agency" where individuals recognize their shared control over outcomes, is vital for AI collectives due to the increasing complexity and impact of their actions in the world (El Zein et al., 2019).

Firstly, AI systems, when acting in concert or in the same space, can have far-reaching effects that single entities may not fully comprehend. A collective sense of agency ensures that these systems are designed and operate within ethical boundaries, acknowledging the consequences of their actions. This is especially important in areas like autonomous driving, healthcare, and financial services, where decisions made by AI collectives can significantly affect human lives.

Secondly, from a legal standpoint, the attribution of responsibility for AI systems and humans is complex. The sense of joint agency among AI systems can help establish clearer lines of responsibility for collective actions within AI but also integrate with existing common sense categories used by humans. Here collaborative awareness should remain compatible with the

so-called network approach to responsibility, where it also becomes possible to identify which systems contributed to a decision and to what extent, thereby facilitating legal accountability.

Thirdly, a collective sense of agency in AI may enhance trust and acceptance among humans. Understanding that AI systems are designed to act with a shared sense of responsibility can reassure the public about their deployment in sensitive areas.

Conclusion

The significance of collaborative awareness lies in its potential to enable safe, efficient, and ethical collaboration across heterogeneous AI agents, thereby accommodating humans' practical and ethical interests in their interactions with groups of AI. By focusing on specific collaborative tasks and dimensions of awareness, this approach offers a measurable and scalable solution to engineering more effective AI systems.

We underscore here the necessity of conceptual engineering in AI research, advocating for the development of bespoke concepts like collaborative awareness that better align with the operational realities and ethical considerations of AI-human interaction. By proposing a dimensional model of collaborative awareness, the project delineates a clear framework for implementing cooperative strategies among AI agents, which can significantly enhance decision-making, efficiency, and adaptability in complex environments.

By advancing beyond traditional models of Central Integrated Awareness (CIA) and Distributed Situational Awareness (DSA), and reusing concepts from biological agents and cooperative games, collaborative awareness proposes a theoretical advancement but also a practical solution to the increasingly complex interactions and coordination challenges posed by diverse and domain-specific AI systems in shared spaces with humans.

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